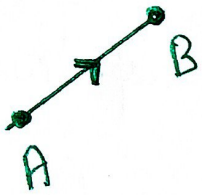
12th BOARD HACKER JUGAADU NOTES:-

Vector:- A vector is represented by a directed line segment. The direction is indicated by the arrow and magnitude by the length of the line segment AB.



A → initial point

B → Terminal point

$\overrightarrow{AB}$



Initial point — Terminal point

$\overrightarrow{AB}$ = Vector $\overrightarrow{AB}$

Magnitude of $\overrightarrow{AB} = |\overrightarrow{AB}|$ = Length of line segment AB

Scalar & Vectors:-

- A Scalar is a quantity which has magnitude but no direction
  e.g: distance
- A vector is a quantity which has magnitude as well as specific direction and obeys the law of addition.
  e.g velocity

Types of Vectors:-

1. Null Vector or Zero Vector:-

If the magnitude of the vector is zero. It can have any arbitrary direction

$|\vec{a}| = 0$

$\Rightarrow \vec{a}$ = Zero Vector of null vector

2.** Unit Vector:-

if the magnitude of a vector is one, then it is called unit vector. $\hat{a}$ is the Unit vector in the direction of $\vec{a}$ and $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$

Unit vector in the direction

of positive x-axis, y-axis & z-axis are $\hat{i}$, $\hat{j}$ and $\hat{k}$

3. Parallel Vectors :-

if the vectors have the same line of support, then they are called parallel vector. The angle between them is $0°$ or $180°$. if $\vec{a}$ and $\vec{b}$ are parallel vectors, then

$$\vec{a} = \lambda \vec{b}, \quad \lambda \in R, \lambda \neq 0$$

i) like parallel vector :-

if angle between vectors is $0°$

$\rightarrow \vec{a}$
$\rightarrow \vec{b}$
} direction of both Vectors is same

ii) Unlike parallel vector :-

if the angle between the vectors are $180°$

$\rightarrow \vec{a}$
$\leftarrow \vec{b}$
} Direction of vectors is just opposite.

**

If $\vec{a}$ & $\vec{b}$ are parallel then

$\vec{a} = \lambda \vec{b}$

λ = Scalar Constant

$\vec{a} = |\vec{a}| \hat{a}$

$|\vec{a}|$ = Magnitude of $\vec{a}$

$\hat{a}$ = Direction of $\vec{a}$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$ → direction of $\vec{a}$

if $|\vec{a}| = 1 \Rightarrow \vec{a}$ is Unit Vector

$\vec{a} = |\vec{a}| \hat{a}$

$\vec{a} = 1 \times \hat{a}$

$\vec{a} = \hat{a}$

$\hat{a}$ = Unit Vector $\Rightarrow |\hat{a}| = 1$

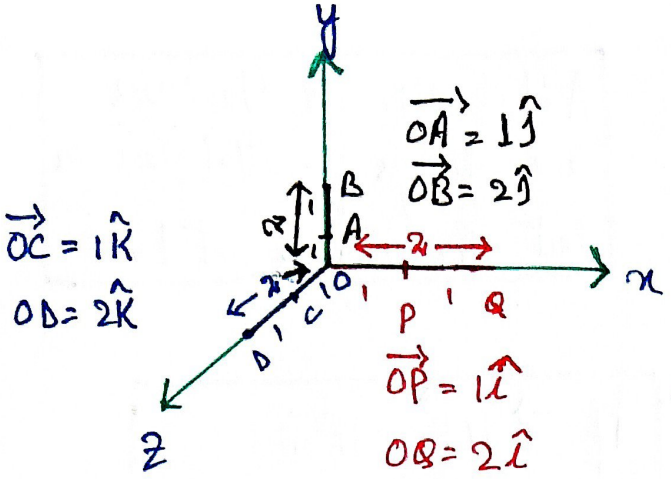
#

$\hat{a}$ is the Unit Vector in the direction of $\vec{a}$.

or

$\hat{a}$ is the Unit Vector that represents the direction of $\vec{a}$

$$|\hat{a}| = 1 \quad \hat{a} = \frac{\vec{a}}{|\vec{a}|}$$



$\hat{i}$ = Unit Vector in the direction of position x-axis

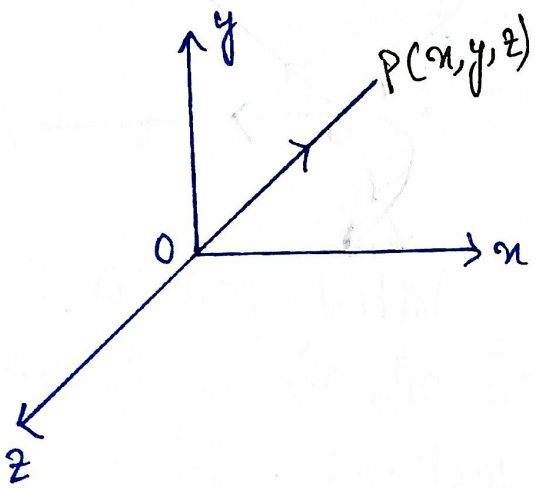
$|\hat{i}| = 1$

$\hat{j}$ = Unit Vector in the direction of position y-axis

$|\hat{j}| = 1$

$\hat{k}$ = Unit Vector in the direction of position z-axis

$|\hat{k}| = 1$



e.g $P(2,3,6)$

$\vec{OP} = 2\hat{i} + 3\hat{j} + 6\hat{k}$

$|\vec{OP}| = \sqrt{(2)^2 + (3)^2 + (6)^2}$

$= \sqrt{4 + 9 + 36}$

$= \sqrt{49}$

$= 7$ Ans.

$\vec{OP}$ = Position Vector of point $P(x,y,z)$

$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

Magnitude of $|\vec{OP}| = \sqrt{x^2 + y^2 + z^2}$

Types of Vectors:-

a) Like parallel Vectors:-

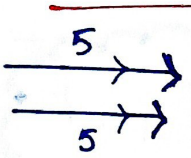
If the angle between the vectors is 0°, then they are called Like Vectors.

b) Unlike parallel Vectors:-

if the angle between two vectors is 180°. then they are called Unlike Vectors.

Types of Vectors:-

④ **Equal Vectors:-** if the magnitude of like parallel vector is same, then they are called equal vectors or the magnitude and the direction of those vectors is same.

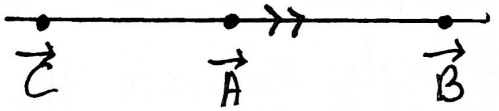


$\vec{a}$ & $\vec{b}$; $|\vec{a}| = |\vec{b}|$ & Direction is same

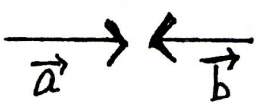
$\Rightarrow \vec{a} = \vec{b}$

⑤. **Collinear Vectors:-**

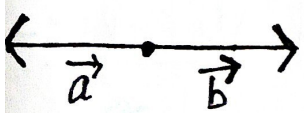
Vectors which are parallel to the same vectors and have either terminal point in common are called colinear vectors.



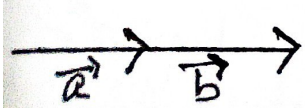
$\vec{AB}$ & $\vec{AC}$ are collinear



Terminal point is common



Initial point is common



Point is common

$\vec{a}$ and $\vec{b}$ are collinear vector.

All collinear Vectors are always parallel vectors but Via Versa is not true

if $\vec{a} \parallel \vec{b}$, then

$\vec{a} = \lambda\vec{b}$

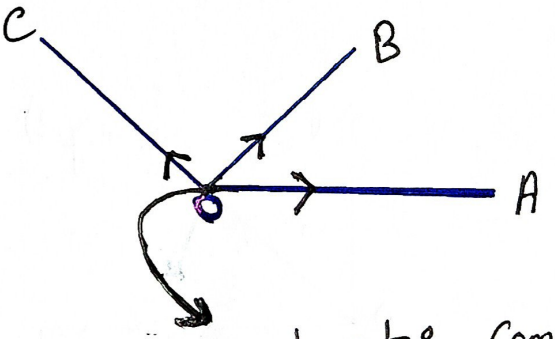
if $\vec{a}$ & $\vec{b}$ are collinear then $\vec{a} = \lambda\vec{b}$

λ = Scalar Constant

Types of Vectors:-

6. **Cointial Vectors:-**

Vectors having same initial point are called cointial Vectors.



Initial point is Common

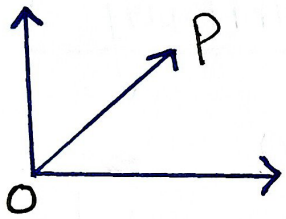
$\vec{OA}$, $\vec{OB}$, $\vec{OC}$ are Coinitial Vectors.

7. Free & Localised Vectors

If at least one of initial & terminal point is fixed, it is called localised vectors otherwise it is free vectors.

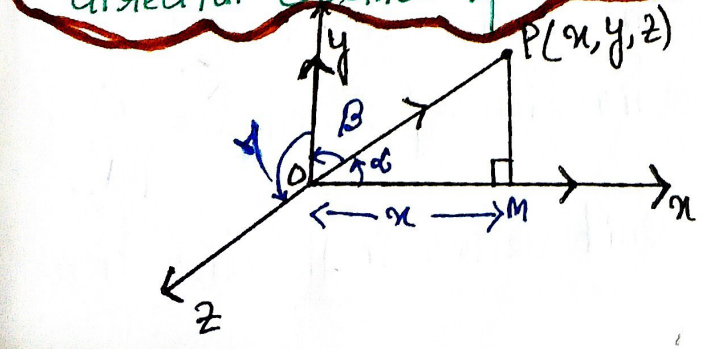
8. Position Vector :-

Position Vector is obtained by joining the point with origin.



- $\vec{OP}$ Position vector of point is $\vec{OP}$
- $\vec{OP}$ position vector of a point is unique

Direction ratios and direction Cosines of Vector



$$OP = |\vec{OP}| = \sqrt{x^2+y^2+z^2}$$

In ΔPMO

$$\cos\alpha = \frac{x}{OP} = \frac{x}{\sqrt{x^2+y^2+z^2}}$$

$$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$$

$\vec{OP}$ make α, β & γ angle with the position direction of x-axis, y-axis & z-axis respectively

$\alpha, \beta, \gamma \rightarrow$ Direction Angles

$\cos\alpha, \cos\beta, \cos\gamma \rightarrow$ Direction Cosines.

$\alpha, \beta, \gamma \in [0, \pi]$ or $[0°, 180°]$

$$l = \cos\alpha = \frac{x}{\sqrt{x^2+y^2+z^2}}$$

$$m = \cos\beta = \frac{y}{\sqrt{x^2+y^2+z^2}}$$

$$n = \cos\gamma = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

Direction Cosines

e.g.

$\overrightarrow{OP} = 2\hat{i} - 3\hat{j} - 6\hat{k}$

$x=2,\ y=-3,\ z=-6$

$\sqrt{x^2+y^2+z^2}$

$= \sqrt{4+9+36}$

$= 7$

$\cos\alpha = \dfrac{x}{\sqrt{x^2+y^2+z^2}} = \dfrac{2}{7}$

$\cos\beta = \dfrac{y}{\sqrt{x^2+y^2+z^2}} = \dfrac{-3}{7}$

$\cos\gamma = \dfrac{z}{\sqrt{x^2+y^2+z^2}} = \dfrac{-6}{7}$

Direction Cosines.

$\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\hat{OP} = \dfrac{\overrightarrow{OP}}{|\overrightarrow{OP}|} \qquad |\hat{OP}| = 1$

$\hat{OP} = \dfrac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2+y^2+z^2}}$

$\hat{OP} = \dfrac{x}{\sqrt{x^2+y^2+z^2}}\hat{i} + \dfrac{y}{\sqrt{x^2+y^2+z^2}}\hat{j} + \dfrac{z}{\sqrt{x^2+y^2+z^2}}\hat{k}$

$\boxed{\hat{OP} = \cos\alpha\hat{i} + \cos\beta\hat{j} + \cos\gamma\hat{k}}$

★★

x- coordinate, y- coordinate & z-coordinate of the Unit Vector are direction Cosines

★★

1) If $\alpha+\beta = \frac{\pi}{2}$, then $\gamma = \frac{\pi}{2}$

2) If $\alpha+\beta > \frac{\pi}{2}$, then 2 possible values of γ are possible

3) If $\alpha+\beta < \frac{\pi}{2}$, then no values of γ is possible.

$\alpha = 30^\circ,\ \beta = 30^\circ$

$\alpha+\beta = 60^\circ < 90^\circ$

$\cos^2 30^\circ + \cos^2 30^\circ + \cos^2\alpha = 1$

$\frac{3}{4} + \frac{3}{4} + \cos^2\alpha = 1$

$\cos^2\alpha = 1 - \frac{6}{4} = \frac{-2}{4}$ (not possible)

α = does not exist

The co-ordinates of the point P may also be expressed as (lr, mr, nr)

The numbers lr, mr & nr, proportional to the direction cosines are called as direction ration of vector $\vec{r}$ & are denoted as a, b and c respectively.

$$\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$$

Direction ratios of $\vec{a} = x, y, z$

** Direction ratios ← Direction cosines are always proportinal

$\vec{a}$ → Direction Ratios = a, b, c

$\vec{a}$ → Direction cosines = l, m, n

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

Q:- Find the direction ratios of $2\hat{i} + 3\hat{j} + 6\hat{k}$ and direction cosines as well

Soln:- Direction Ratios = 2, 3, 6

$a = 2,\ b = 3,\ c = 6$

$\sqrt{4+9+36} = 7$

Direction cosines $= \frac{2}{7}, \frac{3}{7}, \frac{6}{7}$

$l = \frac{2}{7},\ m = \frac{3}{7},\ n = \frac{6}{7}$

$$\frac{l}{a} = \frac{2}{7\times 2} = \frac{1}{7}$$

$$\frac{m}{b} = \frac{3}{7\times 3} = \frac{1}{7}$$

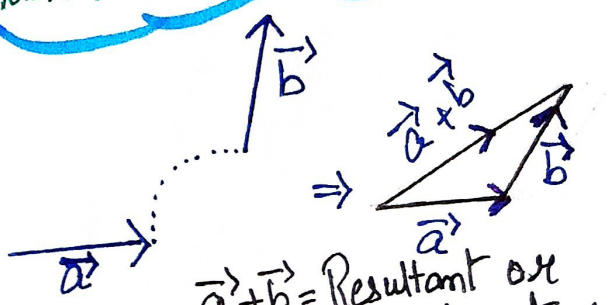
$$\frac{n}{c} = \frac{6}{7\times 6} = \frac{1}{7}$$

$$\Rightarrow \frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

Addition of Vectors :-

- Triangle Law
- Parallelogram Law
- Loop Law

Triangle Law of Vector Addition :-

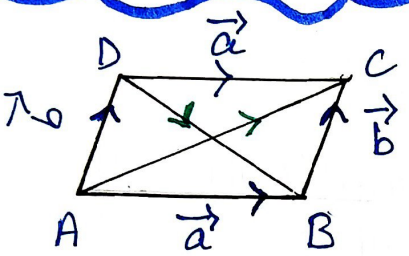


$\vec{a} + \vec{b}$ = Resultant or sum of vectors

If 2 sides of triangle represents $\vec{a}$ & $\vec{b}$, then third side represents $\vec{a}+\vec{b}$



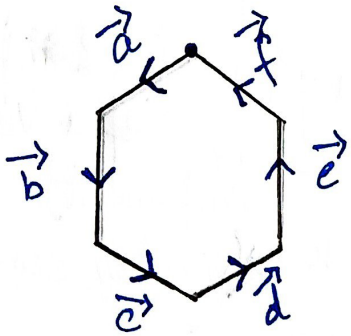
Parallelogram law of Vector Addition



Diagonal $\vec{AC} = \vec{a} + \vec{b}$

Diagonal $\vec{DB} = \vec{a} - \vec{b}$

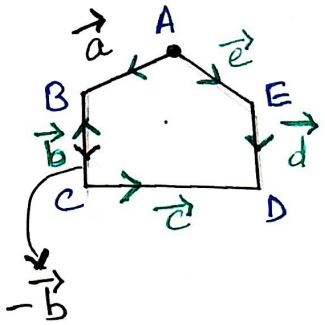
Loop law :-



$\vec{a}+\vec{b}+\vec{c}+\vec{d}+\vec{e}+\vec{f} = 0$

Total Displacement = 0

Eg:-



$\vec{CB} = \vec{b}$

$\vec{BC} = -\vec{b}$

$\vec{AB} + \vec{BC} + \vec{CD} + \vec{DE} + \vec{EA} = 0$

$\vec{a} + (-\vec{b}) + \vec{c} + (-\vec{d}) + (-\vec{e}) = 0$

$\vec{a} - \vec{b} + \vec{c} - \vec{d} - \vec{e} = 0$

$\vec{a} + \vec{c} = \vec{b} + \vec{d} + \vec{e}$

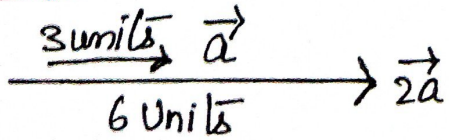
Properties of Vector Addition

1. $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (Commutative)
2. $(\vec{a}+\vec{b}) + \vec{c} = \vec{a} + (\vec{b}+\vec{c})$ (Associative)
3. $\vec{a} + \vec{0} = \vec{a}$ (Additive Identity $\to \vec{0}$)
4. $\vec{a} + -\vec{a} = 0$ (Additive inverse)

$-\vec{a}$ = Addition inverse of $\vec{a}$

Multiplication of a Vector by a Scalar :-

- Let $\vec{a}$ be a given vector & λ scalar then the product of the vector $\vec{a}$ by the scalar λ, denoted as $\lambda\vec{a}$ is called the multiplication of vector $\vec{a}$ by the scalar λ.
- $\lambda\vec{a}$ is also a vector, collinear to the vector $\vec{a}$.
- The vector $\lambda\vec{a}$ has the direction same (or opposite) to that of vector $\vec{a}$ according as the value of λ is positive or negative.
- Magnitude of $\lambda\vec{a}$ is $|\lambda|$ times the magnitude of a vector.



$|\vec{a}| = 3$ units

$2\vec{a}$ = multiplication of $\vec{a}$ by λ constant

$\lambda = 2$

$|2\vec{a}| = |2| \times |\vec{a}|$

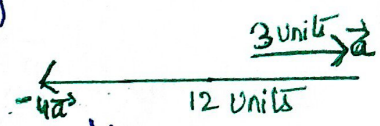
$= 2 \times 3 = 6$ units

$|\lambda\vec{a}| = |\lambda| \times |\vec{a}|$

$|\vec{a}| = 3$ units

$-4\vec{a}$ = multiplication of $\vec{a}$ by -4 (λ)

$\lambda = -4$



$|-4\vec{a}| = |-4| \times |\vec{a}|$

$= 4 \times 3 = 12$

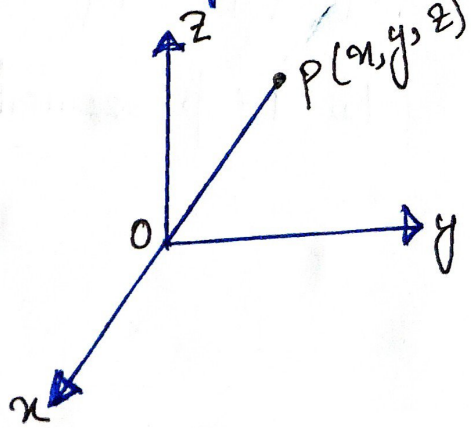
Component of a Vector :-

$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$ (COMPONENT FORM)

$x, y, z \rightarrow$ (Scalar component)

$x\hat{i}, y\hat{j}, z\hat{k} \rightarrow$ (Vector components)

$|\vec{OP}| = r = \sqrt{x^2 + y^2 + z^2}$



If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

1. Sum (or Resultant) of the vectors $\vec{a}$ & $\vec{b}$

$$\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$$

$$\vec{a} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$\vec{b} = -\hat{i} + 4\hat{j} - 5\hat{k}$$

$$\vec{a} + \vec{b} = \hat{i} + 7\hat{j} + \hat{k}$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

2. Difference of the vectors $\vec{a}$ & $\vec{b}$.

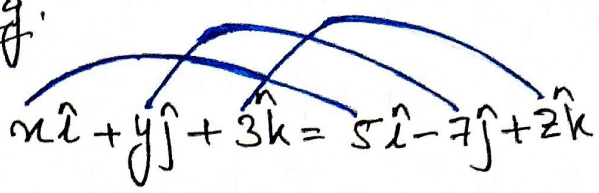
$$\vec{a} - \vec{b} = (a_1 - b_1)\hat{i} + (a_2 - b_2)\hat{j} + (a_3 - b_3)\hat{k}$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

3. Vectors $\vec{a}$ & $\vec{b}$ are equal if & only if

$$a_1 = b_1,\ a_2 = b_2 \text{ & } a_3 = b_3$$

e.g.



$$x\hat{i} + y\hat{j} + 3\hat{k} = 5\hat{i} - 7\hat{j} + z\hat{k}$$

final x, y, z

$$x = 5,\ y = -7,\ z = 3$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

4. Multiplication of $\vec{a}$ by Scalar λ.

$$\lambda\vec{a} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$$

e.g. $\vec{a} = 3\hat{i} - 5\hat{j} + 6\hat{k}$

$$2\vec{a} = 2(3\hat{i} - 5\hat{j} + 6\hat{k})$$

$$= 6\hat{i} - 10\hat{j} + 12\hat{k}$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then -

multiplication of $\vec{a}$ by Scalar λ

$$\lambda\vec{a} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

5. $k\vec{a} + m\vec{a} = (k+m)\vec{a}$

$5\vec{a} + 7\vec{b} + 6\vec{a} - 8\vec{b}$

$11\vec{a} - \vec{b}$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

6. $k(m\vec{a}) = (km)\vec{a}$

$2(5\vec{a}) = 10\vec{a}$

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ & $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

7. $k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}$

$2(\vec{a} + 5\vec{b})$

$= 2\vec{a} + 10\vec{b}$

Remarks:-

1. Two vectors $\vec{a}$ & $\vec{b}$ are parallel or collinear if & only if there exist a non-zero scalar λ such that $\vec{b} = \lambda\vec{a}$

$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$b_1\hat{i} + b_2\hat{j} + b_3\hat{k} = \lambda(a_1\hat{i} + a_2\hat{j} + a_3\hat{k})$

$$\lambda = \frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3}$$

$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

if $\vec{a} \parallel \vec{b}$ or $\vec{a}$ is collinear to $\vec{b}$

$\vec{a} = \lambda\vec{b}$

$a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \lambda(b_1\hat{i} + b_2\hat{j} + b_3\hat{k})$

$a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \lambda b_1\hat{i} + \lambda b_2\hat{j} + \lambda b_3\hat{k}$

$a_1 = \lambda b_1 \quad a_2 = \lambda b_2 \quad a_3 = \lambda b_3$

$\frac{a_1}{b_1} = \lambda \quad \frac{a_2}{b_2} = \lambda \quad \frac{a_3}{b_3} = \lambda$

$\vec{a} \parallel \vec{b}$ or $\vec{a}$ is collinear to $\vec{b}$

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$$

$\Rightarrow$ x-coordinates, y-coordinates and z-coordinates of both the vectors are proportional

Q. Find k if $2\hat{i}-k\hat{j}+7\hat{k}$ & $8\hat{i}-3\hat{j}+28\hat{k}$ are parallel

Soln

$$\frac{2}{8}=\frac{-k}{-3}=\frac{7}{28}=\frac{1}{4}$$

$$\Rightarrow \frac{k}{3}=\frac{1}{4}$$

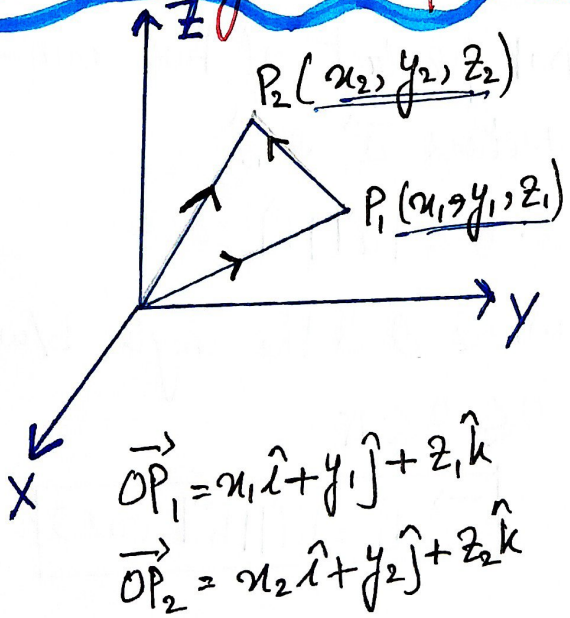
$$\Rightarrow 4k=3$$

$$\boxed{k=\frac{3}{4}}$$

Remarks :-

2. If $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3k$, then a_1, a_2 & a_3 are called direction ratios of $\vec{a}$

Vector Joining Two points :-



$$\vec{OP_1}=x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$$

$$\vec{OP_2}=x_2\hat{i}+y_2\hat{j}+z_2\hat{k}$$

$\vec{P_1P_2}$ = towards P_2

$$\vec{P_1P_2}=\vec{OP_2}-\vec{OP_1}$$

$$=(x_2\hat{i}+y_2\hat{j}+z\hat{k})-(x_1\hat{i}+y_2\hat{j}+z_2\hat{k})$$

$$=(x_2-x_1)\hat{i}+(y_2-y_1)\hat{j}+(z_2-z_1)\hat{k}$$

$$|\vec{P_1P_2}|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

$$\vec{P_1P_2}=\vec{OP_2}-\vec{OP_1}$$

Distance formula :-

$$\vec{OA}=x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$$ ✓

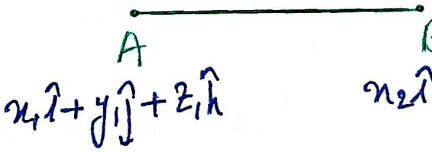
$$\vec{OB}=x_2\hat{i}+y_2\hat{j}+z_2\hat{k}$$ ✓

$$\vec{AB}=\vec{OB}-\vec{OA}$$ ✓

$$=x_2\hat{i}+y_2\hat{j}+z_2\hat{k}-(x_1\hat{i}+y_1\hat{j}+z_1\hat{k})$$

$$=(x_2-x_1)\hat{i}+(y_2-y_1)\hat{j}+(z_2-z_1)\hat{k}$$

$$|\vec{AB}|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$



$x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$ $\quad$ $x_2\hat{i}+y_2\hat{j}+z_2\hat{k}$

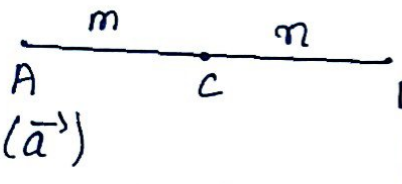
$$\vec{AB}=(x_2-x_1)\hat{i}+(y_2-y_1)\hat{j}+(z_2-z_1)\hat{k}$$

$$|\vec{AB}|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

Section formula :-

A, B & C are collinear

Case 1 :- C divides the line segment joining A & B internally in the ratio m:n.

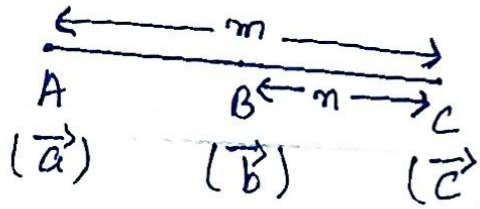


$$\frac{AC}{BC} = \frac{m}{n} \Rightarrow AC : BC = m : n$$

$$C\left[\frac{m\vec{b} + n\vec{a}}{m+n}\right]$$

Case 2 :-

C divides AB externally in the ratio m:n



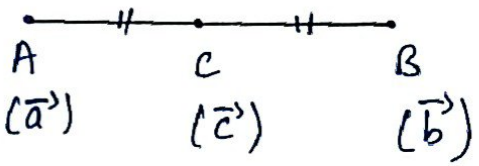
C does not lie b/w A & B

$$\frac{AC}{BC} = \frac{m}{n}$$

$$\boxed{\vec{c} = \frac{m\vec{b} - n\vec{a}}{m-n}}$$

Case 3 :-

C is the midpoint of AB



$$\boxed{\vec{c} = \frac{\vec{a} + \vec{b}}{2}}$$

Product of two Vectors :-

- Scalar or Dot product
- Vector or Cross product

Scalar (or dot) product of two Vectors :-

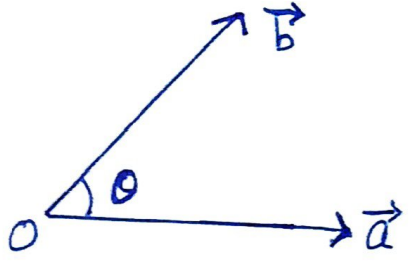
Dot product of two non-zero vectors $\vec{a}$ & $\vec{b}$

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$

where θ is the angle b/w $\vec{a}$ & $\vec{b}$

$0 \leq \theta \leq \pi$

$$\boxed{\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta}$$



$0 \leq \theta \leq \pi$

e.g. If $|\vec{a}| = 4$, $|\vec{b}| = 3$, $\theta = \pi/3$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \pi/3$

$= 4 \times 3 \times \frac{1}{2}$

$= 6$ Ans.

Observations :-

1. $\vec{a} \cdot \vec{b}$ is a real number (Scalar quantity)

** 2. Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors, then $\vec{a} \cdot \vec{b} = 0$ if and only if $\vec{a}$ and $\vec{b}$ are perpendicular to each other i.e

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$\vec{a} \cdot \vec{b} = 0$

$|\vec{a}||\vec{b}| \cos\theta = 0$

$\cos\theta = 0$

$\theta = \frac{\pi}{2} \Rightarrow \vec{a} \perp \vec{b}$

3. If $\theta = 0$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$

In particular, $\vec{a} \cdot \vec{a} = |\vec{a}|^2$, as θ in this case 0.

4. If $\theta = \pi$, then $\vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$

In particular, $\vec{a}(\vec{a})$ $|\vec{a}|^2$, as θ in this case is π.

$\theta = 0^\circ$

① $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos 0^\circ$

$= |\vec{a}||\vec{b}| \times 1$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$

2. $\theta = \pi$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\pi$

$= |\vec{a}||\vec{b}| \times -1$

$= -|\vec{a}||\vec{b}|$

* $\vec{a} \cdot \vec{a} = |\vec{a}||\vec{a}| \cos 0^\circ$

$= |\vec{a}||\vec{a}| \times 1$

$= |\vec{a}|^2$

$$\boxed{\vec{a} \cdot \vec{a} = |\vec{a}|^2}$$

$(\vec{a}+\vec{b}) \cdot (\vec{a}+\vec{b}) = |\vec{a}+\vec{b}|^2$

$(\vec{a}-\vec{b}) \cdot (\vec{a}-\vec{b}) = |\vec{a}-\vec{b}|^2$

$(\vec{a}+\vec{b}+\vec{c}) \cdot (\vec{a}+\vec{b}+\vec{c}) = |\vec{a}+\vec{b}+\vec{c}|^2$

Observation:-

5. In view of the observations 2 and 3, for mutually perpendicular unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$, we have

$$\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$$
$$\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$$

6. The angle b/w two non-zero vectors $\vec{a}$ and $\vec{b}$ is given by

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} \text{ or } \theta = \cos^{-1}\left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right)$$

7. The Scalar product is commutative i.e

$$\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$$

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$$

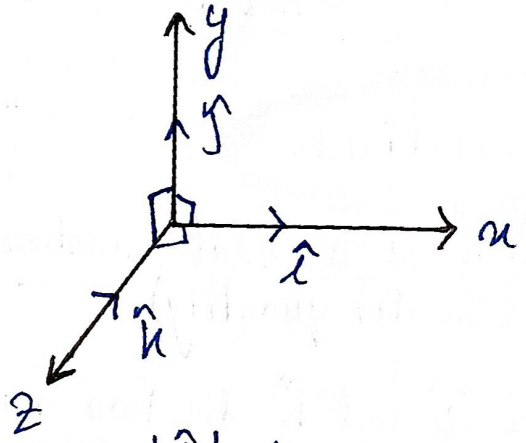
*It is used to calculate the angle b/w 2 vectors.

$$\hat{i}\cdot\hat{i} = |\hat{i}||\hat{i}|\times\cos\theta^\circ$$
$$1\times1\times1$$
$$\hat{i}\cdot\hat{i} = 1$$
$$\hat{j}\cdot\hat{j} = 1$$
$$\hat{k}\cdot\hat{k} = 1$$

$$\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$$

$$\hat{i}\cdot\hat{j} = |\hat{i}||\hat{j}|\cos 90^\circ$$
$$\hat{i}\cdot\hat{j} = 0$$
$$\hat{j}\cdot\hat{k} = |\hat{j}||\hat{k}|\cos 90^\circ$$
$$\hat{j}\cdot\hat{k} = 0$$
$$\hat{k}\cdot\hat{i} = |\hat{k}||\hat{i}|\cos 90^\circ$$
$$\hat{k}\cdot\hat{i} = 0$$

$$\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$$



$$|\hat{i}| = 1$$
$$|\hat{j}| = 1$$
$$|\hat{k}| = 1$$

***

1) $(\vec{a}+\vec{b})^2 = (\vec{a}+\vec{b})(\vec{a}+\vec{b}) = |\vec{a}+\vec{b}|^2$
$= \vec{a}(\vec{a}+\vec{b}) + \vec{b}\cdot(\vec{a}+\vec{b})$
$= \vec{a}\cdot\vec{a} + \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{a} + \vec{b}\cdot\vec{b}$
$= |\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + |\vec{b}|^2$

1. $(\vec{a}+\vec{b})^2 = (\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b}) = |\vec{a}+\vec{b}|^2 = |a|^2 + |b|^2 + 2\vec{a}\cdot\vec{b}$

2. $(\vec{a}-\vec{b})^2 = (\vec{a}-\vec{b})(\vec{a}-\vec{b}) = |\vec{a}-\vec{b}|^2 = |a|^2+|b|^2 - 2\vec{a}\cdot\vec{b}$

3. $(\vec{a}+\vec{b}).(\vec{a}-\vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$

4. $(\vec{a}+\vec{b}+\vec{c})^2 = (\vec{a})^2+(\vec{b})^2+(\vec{c})^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$
$|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$

Two Important properties of Scalar product

Property 1.

Distributivity of scalar product over addition) Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be any three vectors, then

$$\vec{a}\cdot(\vec{b}+\vec{c}) = \vec{a}\cdot\vec{b}+\vec{a}+\vec{c}$$

Property 2.

$$\vec{a}\cdot\vec{b} = a_1b_1+a_2b_2+a_3b_3$$

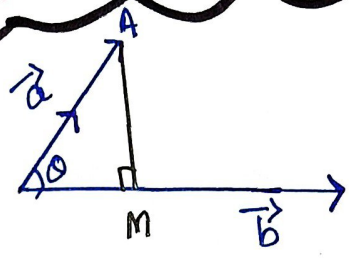
e.g $\vec{a} = 2\hat{i}-3\hat{j}+4\hat{k}$
$\vec{b} = -\hat{i}+\hat{j}-5\hat{k}$

$\vec{a}\cdot\vec{b} = (2\times-1)+(-3\times1)+(4\times-5)$
$= -2-3-20 = -25$ Ans

$\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$
$\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

$$\vec{a}\cdot\vec{b} = a_1b_1+a_2b_2+a_3b_3$$

Projection and protection vector :-



OM = length of projection of $\vec{a}$ on $\vec{b}$

$OA = |\vec{a}|$

In ΔAMO

$\cos\theta = \frac{OM}{OA}$

$OM = OA\cos\theta$

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$$

$OM = OA\cos\theta$
$= |\vec{a}| \times \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}\right)$

$OM = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \vec{a}\cdot\frac{\vec{b}}{|\vec{b}|} = \vec{a}\cdot\hat{b}$

Projection of $\vec{a}$ on $\vec{b}$ $= \vec{a}\cdot\hat{b}$

Projection of $\vec{b}$ on $\vec{a}$ $= \vec{b}\cdot\hat{a}$

Direction $\overrightarrow{OM}$ = Direction of $\vec{b}$

$= \hat{b}$

Projection Vector $= \overrightarrow{OM} = |\overrightarrow{OM}|\,\hat{b}$

$= (OM)\hat{b} = (\vec{a}\cdot\hat{b})\hat{b}$

Vector (or cross) product of two Vectors :-

The vector product of two non-zero vectors $\vec{a}$ & $\vec{b}$ is denoted by $\vec{a} \times \vec{b}$ & defined as

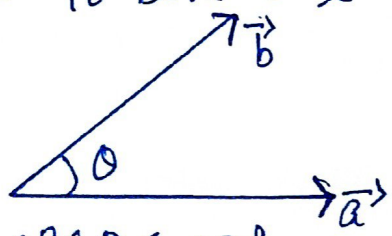
$$\vec{a}\times\vec{b} = |\vec{a}|\,|\vec{b}|\sin\theta\,\hat{n}$$

where θ is the angle b/w $\vec{a}$ & $\vec{b}$, $0 \le \theta \pi$ & $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ & $\vec{b}$

$$\vec{a}\times\vec{b} = |\vec{a}|\,|\vec{b}|\sin\theta\,\hat{n}$$

Magnitude of $\vec{a}\times\vec{b} = |\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$

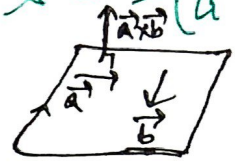
Direction of $\vec{a}\times\vec{b}\times\hat{n}$ = Unit vector which is $\perp$ to both $\vec{a}$ & $\vec{b}$



$(\vec{a}\times\vec{b}) \perp \vec{a}$ & $(\vec{a}\times\vec{b}) \perp \vec{b}$

** Direction of Vector $\perp$ to both $\vec{a}$ & $\vec{b}$ = Direct^n of $(\vec{a}\times\vec{b})$

** Vector which is $\perp$ to both $\vec{a}$ & $\vec{b}$ $= (\vec{a}\times\vec{b})$



Unit vector which is $\perp$ to both $\vec{a}$ & $\vec{b}$

$$= 1 \times \left(\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}\right) = \frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$$

Observation :-

1. $\vec{a}\times\vec{b}$ is a vector. ✓

2. Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors. Then $\vec{a}\times\vec{b} = 0$ if and only if $\vec{a}$ and $\vec{b}$ are parallel (or collinear) to each other i.e

$$\vec{a}\times\vec{b} = \vec{0} \Leftrightarrow \vec{a} \parallel \vec{b}$$

$\vec{a}\times\vec{b} = \vec{0}$

$|\vec{a}\times\vec{b}| = 0$

$|\vec{a}|\,|\vec{b}|\sin\theta = 0$

$\sin\theta = 0$

$\theta = 0$ or 180°

$\vec{a} \parallel \vec{b}$

In Particular, $\vec{a} \times \vec{a} = \vec{0}$ and $\vec{a} \times (-\vec{a}) = \vec{0}$, Since in the first situation, $\theta = 0$ and in the 2nd one, $\theta = \pi$, making the value of $\sin\theta$ to be 0.

4. In view of observations 2,3 for, mutually $\perp$ unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$ (figure), we have

$$\boxed{\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}}$$

$$\boxed{\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}}$$

5. In terms of vector product, the angle b/w two vectors $\vec{a}$ and $\vec{b}$ may be given as

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}|\,|\vec{b}|}$$

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$$

$$\boxed{\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}}$$

$$\hat{i} \times \hat{i} = |\hat{i}||\hat{i}| \sin\theta$$
$$= 1 \times 1 \times 0 = 0$$

$$\hat{i} \times \hat{i} = 0$$
$$\hat{j} \times \hat{j} = 0$$
$$\hat{k} \times \hat{k} = 0$$

$$\boxed{\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0}$$

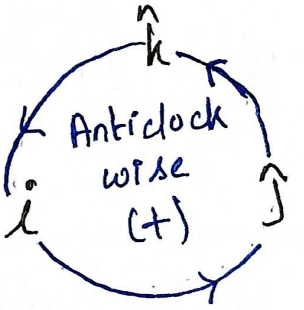
$$\hat{i} \times \hat{j} = |\hat{i}||\hat{j}| \sin 90^\circ \hat{k}$$
$$= 1 \times 1 \times 1 \times \hat{k}$$

$$\boxed{\hat{i} \times \hat{j} = k}$$

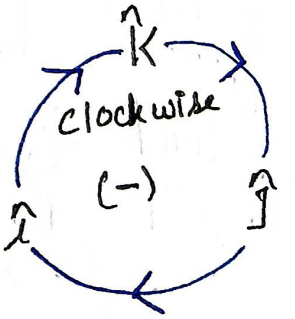
$$\hat{j} \times \hat{k} = |\hat{j}||\hat{k}| \sin 90^\circ \hat{i}$$
$$= 1 \times 1 \times 1 \times \hat{i}$$

$$\boxed{\hat{j} \times \hat{k} = \hat{i}}$$

$$\boxed{\hat{k} \times \hat{i} = \hat{j}}$$



$\hat{i} \times \hat{j} = k$
$\hat{j} \times \hat{k} = \hat{i}$
$\hat{k} \times \hat{i} = \hat{j}$



$\hat{i} \times \hat{k} = -\hat{j}$
$\hat{k} \times \hat{j} = -\hat{i}$
$\hat{j} \times \hat{i} = -\hat{k}$

Observations :-

6. Vector product is not commutative

$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$

$\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$

7. $\hat{j} \times \hat{i} = -\hat{k}$, $\hat{k} \times \hat{j} = -\hat{i}$ and $\hat{i} \times \hat{k} = -\hat{j}$

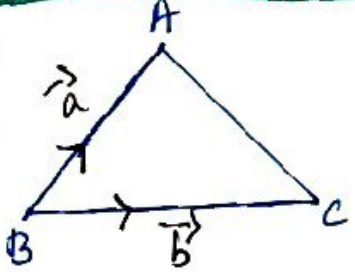
$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ ✓

$\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ ✓

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \ldots$$

$= \hat{i}[a_2b_3 - a_3b_2] - \hat{j}[a_1b_3 - a_3b_1] + \hat{k}[a_1b_2 - a_2b_1]$

Area of the triangle :-



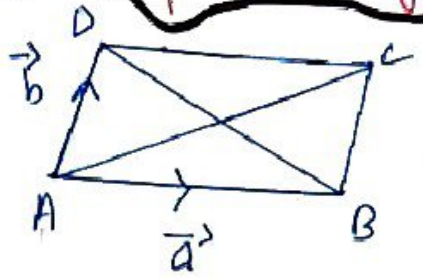
$\vec{a} = |\vec{BA}|$

$\vec{b} = |\vec{BC}|$

$\vec{a}$ & $\vec{b}$ represents the adjacent sides of the triangle, then

Area of $\Delta ABC = \frac{1}{2}|\vec{a} \times \vec{b}|$

Area of Parallelogram



$\vec{a}$ & $\vec{b}$ represents the adjacent sides of the llgm ABCD, then

Area of llgm $= |\vec{a} \times \vec{b}|$

if $\vec{d_1}$ & $\vec{d_2}$ represents the diagonals vectors of the llgm

$\vec{d_1} = \vec{AC}$ $\quad \vec{d_2} = \vec{DB}$

Area of llgm $= \frac{1}{2}|\vec{d_1} \times \vec{d_2}|$